

We show that σ' is bijective on $\mathbf{GF}(2^m)$. Let us fix $u, y_1, y_2 \in \mathbf{GF}(2^m)$, $y_1 \neq y_2$. There is an element $v \in \mathbf{GF}(2^t)$ such that

$$\mathrm{Tr}_{\mathbf{GF}(2^t)/\mathbf{GF}(2)}(uv) \neq h'(y_1) + h'(y_2).$$

As τ is surjective, there is a $z \in \mathbf{GF}(2^m)$ with $\tau(z) = v$. By Lemma 7, there is $x \in \mathbf{GF}(2^m)$ with $y_1 \star x + y_2 \star x = z$. Then

$$\begin{aligned} \alpha_{y_1}(x) + \alpha_{y_2}(x) &= \mathrm{Tr}_{\mathbf{GF}(2^t)/\mathbf{GF}(2)}(u\tau(y_1 \star x + y_2 \star x) + uh(y_1) + uh(y_2)) \\ &= \mathrm{Tr}_{\mathbf{GF}(2^t)/\mathbf{GF}(2)}(u\tau(z)) + h'(y_1) + h'(y_2) \\ &= \mathrm{Tr}_{\mathbf{GF}(2^t)/\mathbf{GF}(2)}(uv) + h'(y_1) + h'(y_2) \\ &\neq 0. \end{aligned}$$

This implies $\alpha_{y_1}(x) \neq \alpha_{y_2}(x)$ and $\sigma'(y_1) \neq \sigma'(y_2)$. It follows that all component functions of f are of Maiorana-McFarland type bent function. This finishes the proof of the theorem. $\square$

In this paper, we do not study the question of EA-equivalence of the $(2m, t)$ -bent functions defined above. In general, this is a very difficult question. We only remark that Weng, Feng and Qui [23] proved that most of the $\mathcal{PS}$ type bent functions, obtained from a Desarguesian spread are not EA-equivalent to any Maiorana-McFarland bent function. This leads us to conclude that, typically, for $t > 1$, the $(2m, t)$ -bent functions described by (9) are generally not EA-equivalent to the other two classes, as specified in (6) and (8).

3.4. Proof of Theorem 1. Let us recall the Carlet-Ding-Yuan bound (2) for the distance between affine and (n, m) -bent functions:

$$\left(1 - \frac{1}{2^m}\right) (2^n - 2^{n/2}) \leq d_H(f, \mathcal{A}) \leq \left(1 - \frac{1}{2^m}\right) (2^n + 2^{n/2}).$$

For an (n, m) -bent function f , the Walsh coefficients are

$$W_f(a, b) = \begin{cases} \pm 2^{n/2} & \text{if } b \neq 0, \\ 0 & \text{if } a \neq 0, b = 0, \\ 2^n & \text{if } a = 0, b = 0. \end{cases}$$

Hence, the Carlet-Ding-Yuan bound follows from Lemma 3 easily. The Liu-Mesnager-Chen Conjecture implies that the true value of $d_H(f, \mathcal{A})$ is $(1 - \frac{1}{2^m}) (2^n - 2^{n/2})$. Theorem 1 claims that this holds for two classes of (n, m) -bent functions.

Proof of Theorem 1. Let E_i be the set of pairs $(x, y) \in \mathbf{GF}(2^m)^2$ such that $f_i(x, y) = f_i(0, 0)$, $i = 1, 2$. We show that $|E_i| = 2^{2m-t} + 2^m - 2^{m-t}$, which implies that f_i has Hamming distance $(1 - 2^{-t})(2^{2m} - 2^m)$ from the constant function $f_i(0, 0)$. Therefore, $d_H(f_i, \mathcal{A}) \leq (1 - 2^{-t})(2^{2m} - 2^m)$, and the theorem follows from the Carlet-Ding-Yuan bound.

Let T be the set of elements $z \in \mathbf{GF}(q^m)$ with $\gamma(z) = f_1(0, 0)$. Since γ is balanced, $|T| = 2^{m-t}$, and $f_1(x, y) = f_1(0, 0)$ if and only if $\star_{\frac{x}{y}} \in T$. Moreover, since $\star_{\frac{0}{0}} = 0$, we have $0 \in T$. The number of solutions of $\star_{\frac{x}{y}} = 0$ is $2 \cdot 2^m - 1$, and the number of solutions of $\star_{\frac{x}{y}} = t \in T \setminus \{0\}$ is $2^m - 1$. This implies

$$|E_1| = |f_1^{-1}(f_1(0, 0))| = 2 \cdot 2^m - 1 + (2^{m-t} - 1)(2^m - 1).$$

The same argument applies to f_2 . In this case, $f_2(0, 0) = h(0)$, and the number of solutions of $\sigma(y) \star x = t$ is $2 \cdot 2^m - 1$ or $2^m - 1$, depending on $t = 0$ or $t \neq 0$. $\square$